

Sex-linked genes exhibit unique patterns of inheritance

Morgan's discovery of a trait (white eyes) that correlated with the sex of flies provided important support for the chromosome theory of inheritance. Because the identity of the sex chromosomes in an individual could be inferred by observing the sex of the fly, the behavior of the two members of the pair of sex chromosomes could be correlated with the behavior of the two alleles of the eye color gene. In this section, we'll take a closer look at the role of sex chromosomes in inheritance.

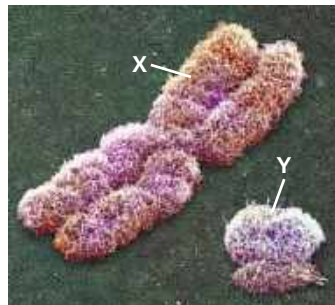
The Chromosomal Basis of Sex

Although sex has traditionally been described as binary—male or female—we are coming to understand that this classification may be too simplistic. Here, we use the term *sex* to refer to classification into a group with a shared set of anatomical and physiological traits. In this sense, sex in many species is determined largely by inheritance of sex chromosomes. (The term *gender*, previously used as a synonym of sex, is now more often used to refer to an individual's own experience of identifying as male, female, or otherwise.)

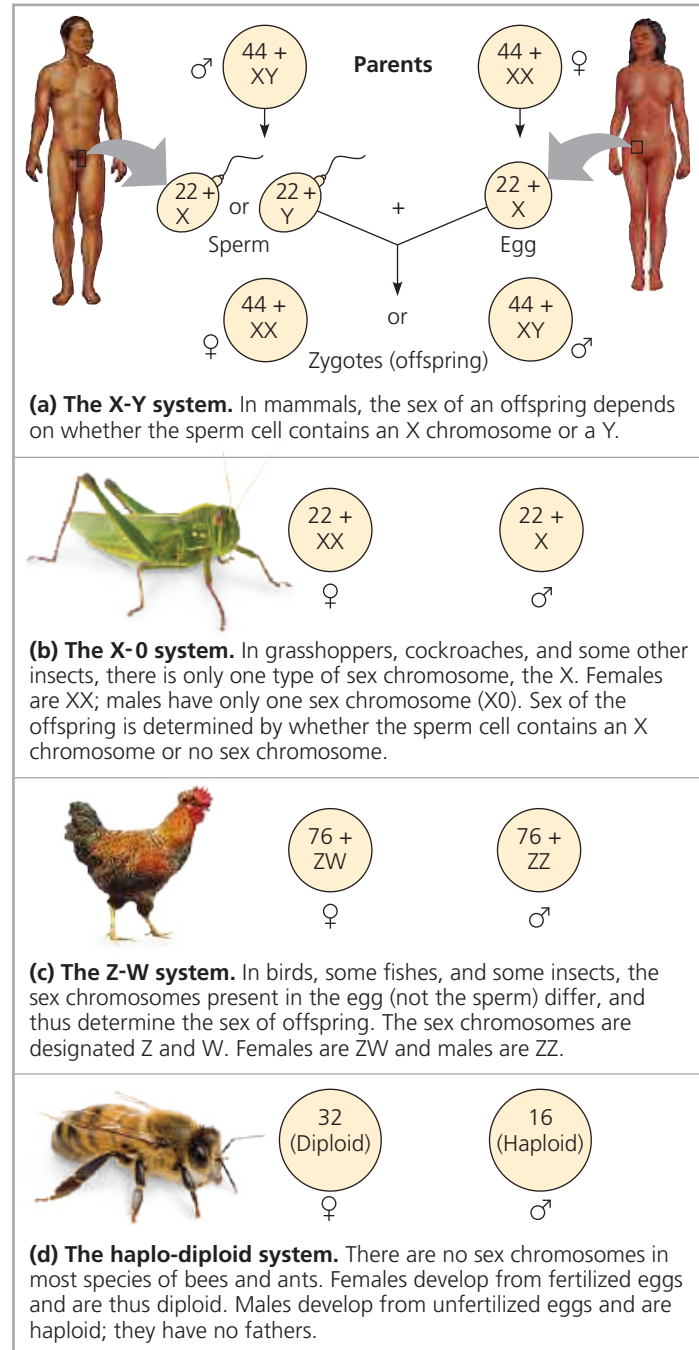
Humans and other mammals have two types of sex chromosomes, designated X and Y. The Y chromosome is much smaller than the X chromosome (Figure 15.5). A person who inherits two X chromosomes, one from each parent, usually develops anatomy we associate with the “female” sex, while “male” properties are associated with the inheritance of one X chromosome and one Y chromosome (Figure 15.6a). Short segments at either end of the Y chromosome are the only regions that are homologous with regions on the X. These homologous regions allow the X and Y chromosomes in males to pair and behave like homologs during meiosis in the testes.

In mammalian testes and ovaries, the two sex chromosomes segregate during meiosis. Each egg receives one X chromosome. In contrast, sperm fall into two categories: Half the sperm cells a male produces receive an X chromosome, and half receive a Y chromosome. We can trace the sex of each offspring to the events of conception: If a sperm cell bearing an X chromosome fertilizes an egg, the zygote is XX, a female; if a sperm cell containing a Y chromosome fertilizes an egg, the zygote is XY, a male (see Figure 15.6a). Thus, in general, sex determination is a matter of chance—a fifty-fifty chance. Note that the mammalian X-Y system isn't the only chromosomal system for determining sex. Figure 15.6b–d illustrates three other systems.

▼ Figure 15.5 Human sex chromosomes (duplicated).



▼ **Figure 15.6 Some chromosomal systems of sex determination.** Numerals indicate the number of autosomes in the species pictured. In *Drosophila*, males are XY, but sex depends on the ratio of the number of X chromosomes to the number of autosomes, not simply on the presence of a Y chromosome. In some species (not shown here), sex is determined not by chromosomes but by environmental factors such as temperature.



In humans, the anatomical signs of sex begin to emerge when the embryo is about 2 months old. Before then, the rudiments of the gonads are generic—they can develop into either testes or ovaries, depending on whether or not a Y chromosome is present, and depending on what genes are active. A gene on the Y chromosome—called *SRY*, for *sex-determining region of Y*—is required for the development of testes. In the absence of *SRY*, the gonads develop into ovaries, even in an XY embryo.